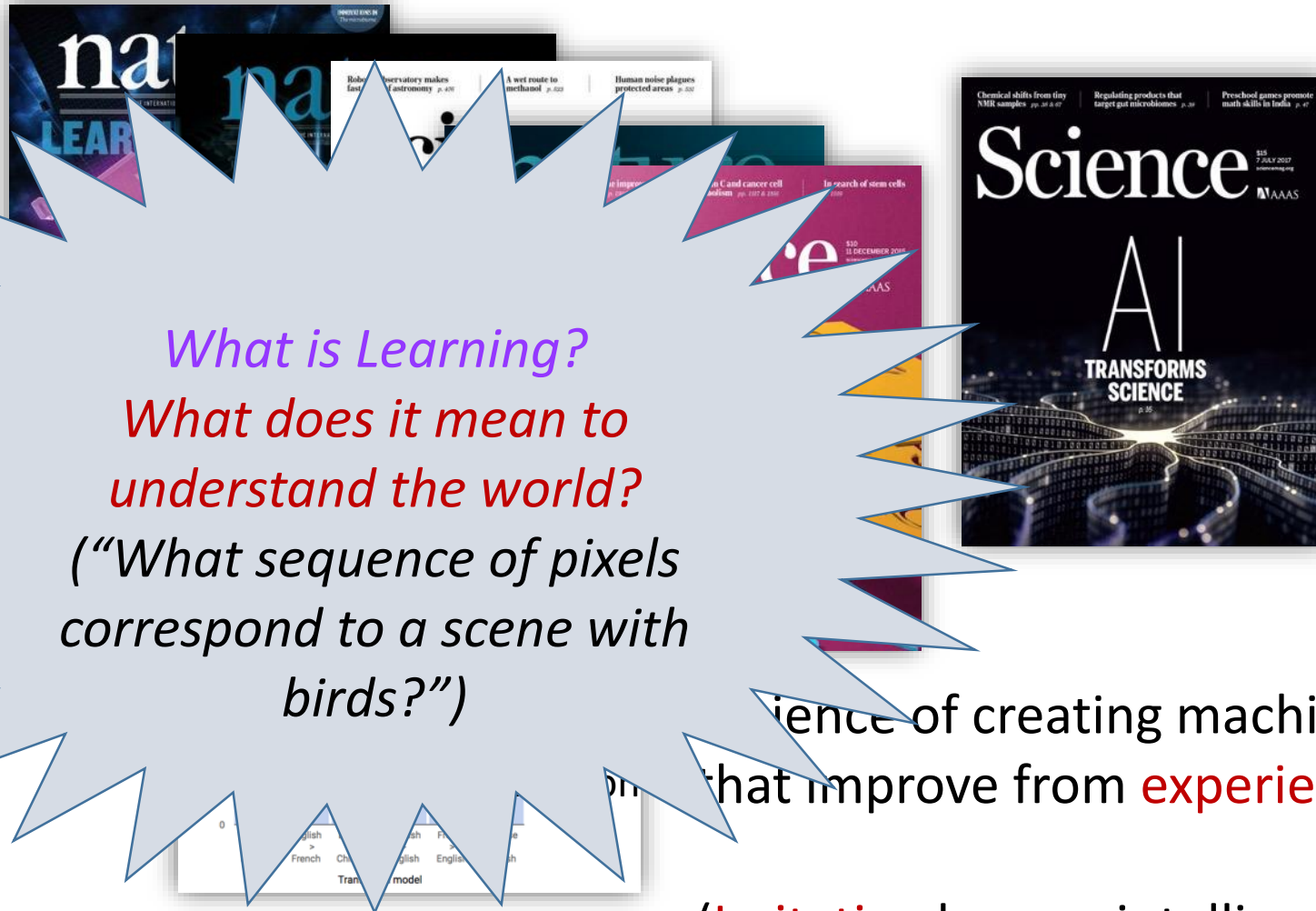


Machine Learning

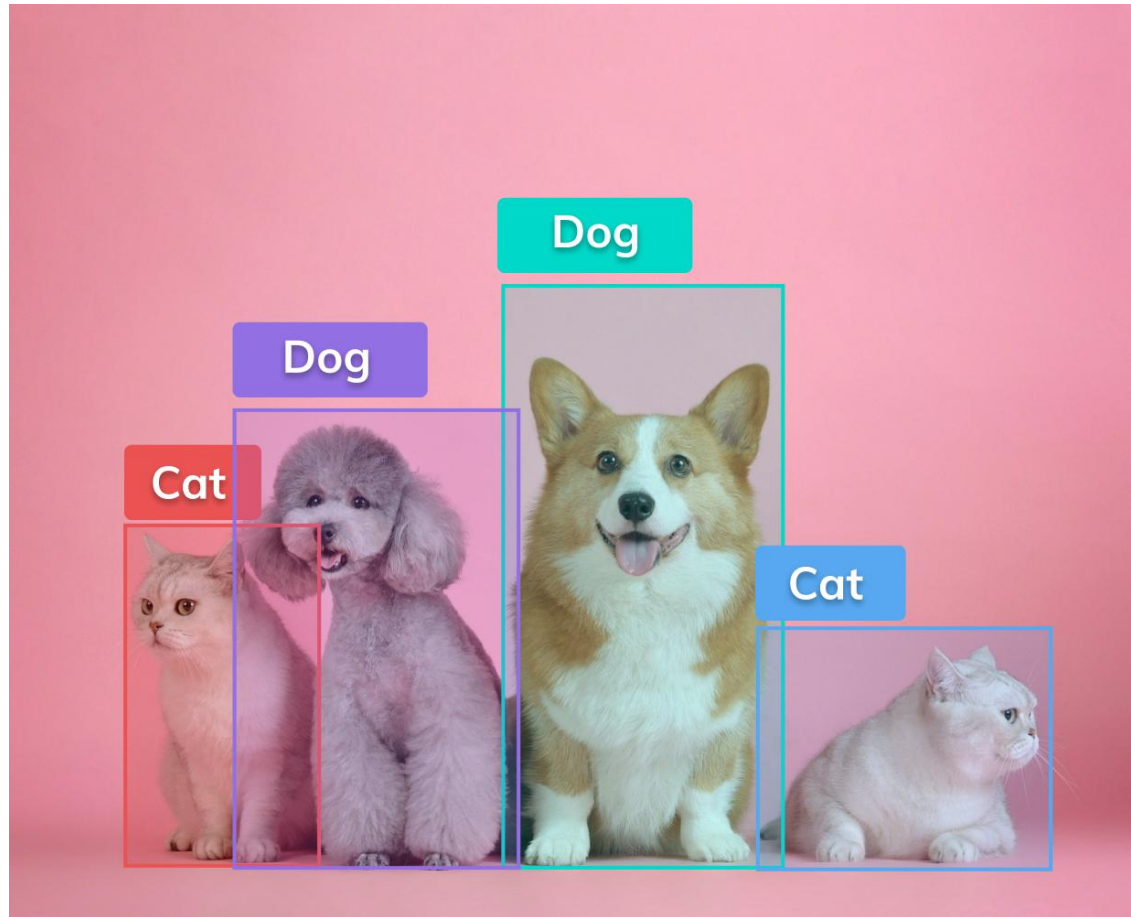
A new kind of Science

Machine Learning (ML): A new kind of science



science of creating machines/programs
that improve from **experience** and **interaction**."
(**Imitating** human intelligence not an explicit goal.)

Applications: Object Detection

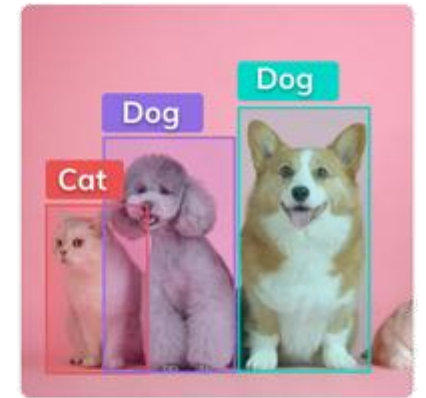


Classification



Cat

Detection



Cat, Dog, Dog

Applications: Tracking



SORT result on MOT15 dataset

Application: Surveillance



- Recognizing a person when the person is not close to the image acquisition device.



Scface dataset

Applications: Emotion Recognition



Applications: Anti-spoofing



(a)



(b)



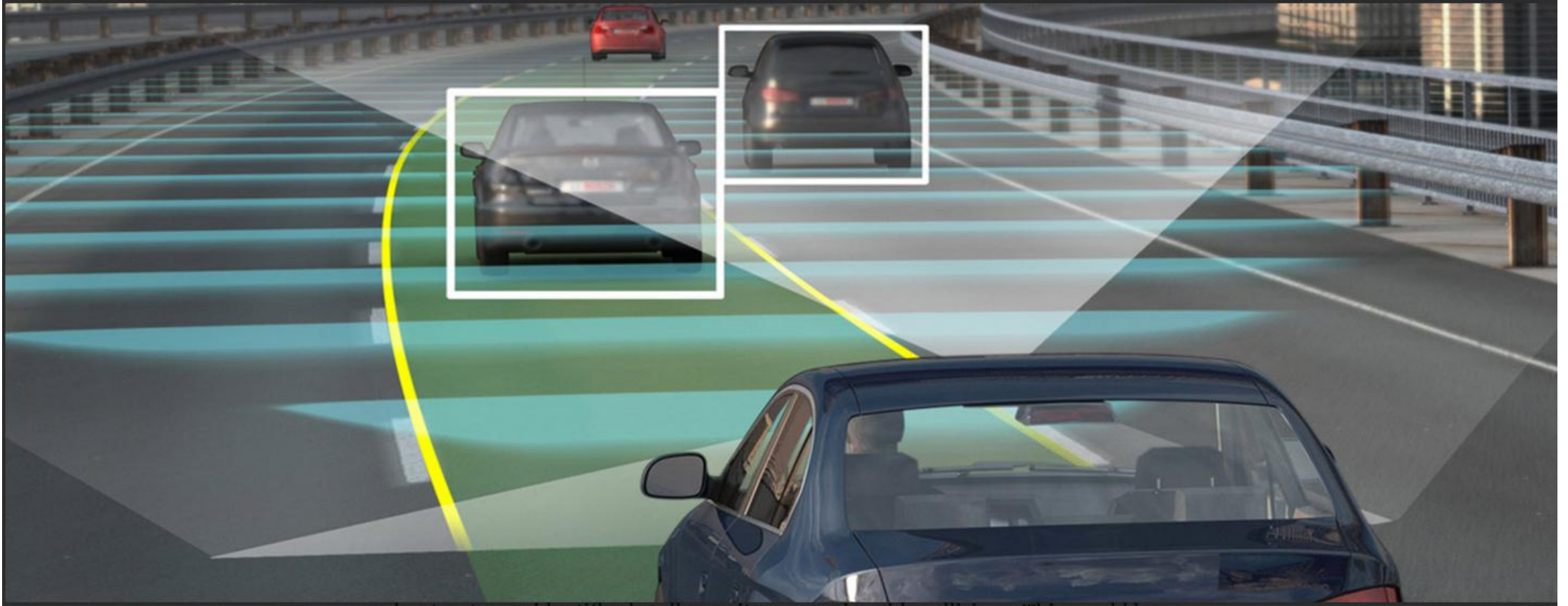
(c)



(d)

(a) real face (b) spoof attack by using printed photograph (c) spoof attack by using printed photograph with cut eyes (d) spoof attack using video from CASIA-FASD database.

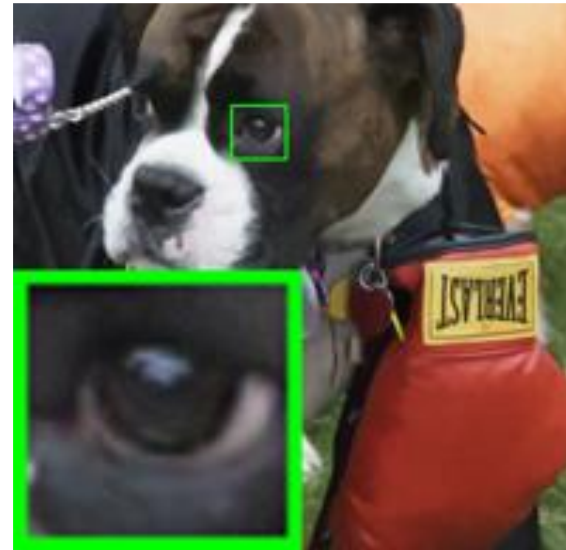
Applications: Self-driving Car



ADAS- Cruise Control, Emergency Braking, Lane Assistance, Collision warning, Traffic Sign Detection, Parking

Image source: geospatialworld

Applications: Denoising



Applications: Super-Resolution



Input LR video (180×256) – Resized



Output HR video (540×768) – Resized

Applications: Dehazing



Applications: Image Colorization

- Objective is to colorize a gray-scale image.

